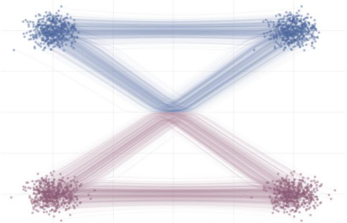
FlowRL: Flow-Balanced Policy Optimization for LLM Reasoning

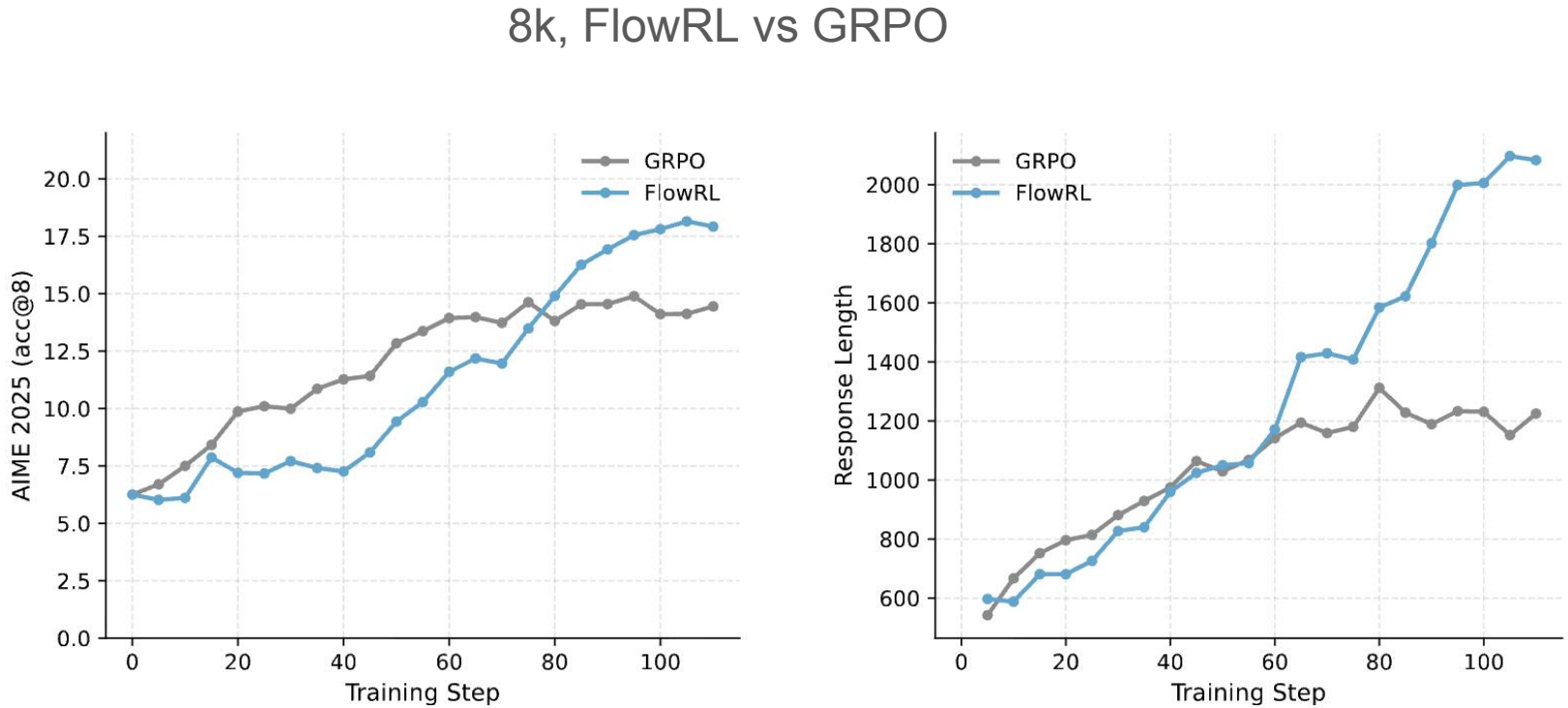


Flow in = Flow out

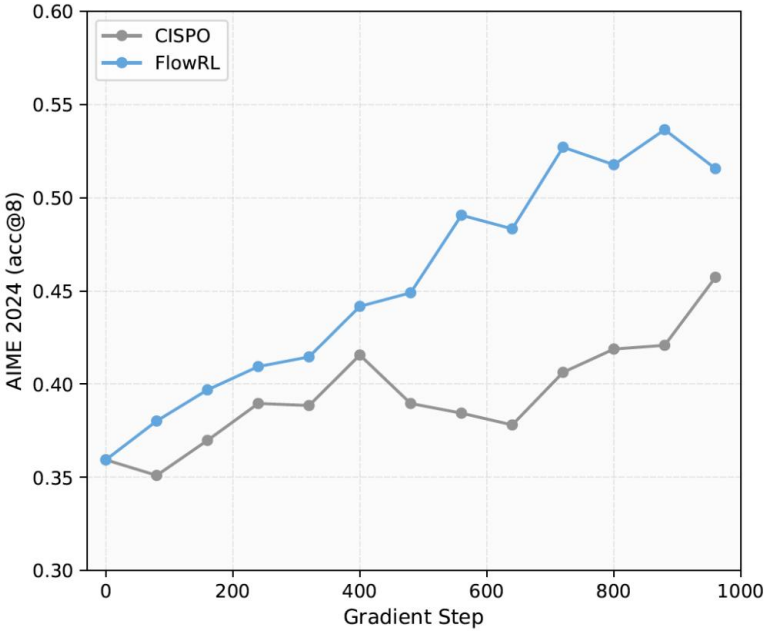
Enable Efficient Exploration and Diverse Reasoning

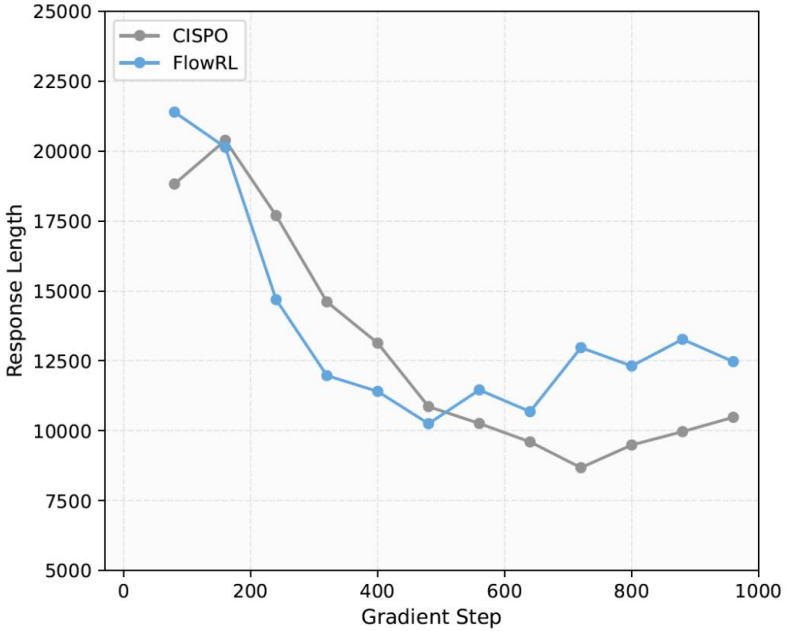
Xuekai Zhu

Main Results



21k, FlowRL vs CISPO

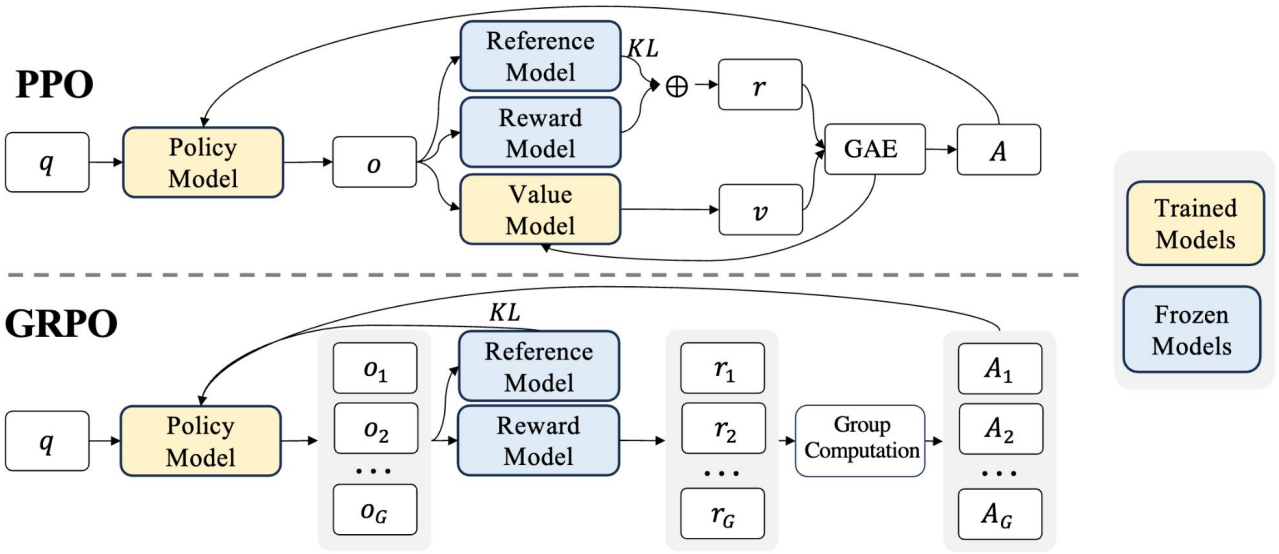




Outline

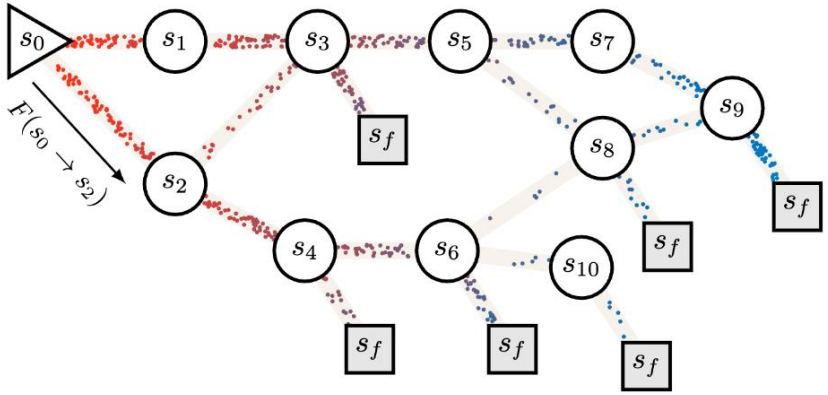
- Motivation
  - Reward Maximization vs. Distribution Matching
- Preliminaries
  - REINFORCE, PPO, GRPO
  - GFlowNets
- Q1: How to Shift Reward Maximization to Distribution Matching ?
  - KL Objective
  - Trajectory Balance
- Q2: How to Ensure Training Stability for Flow-Balanced Objective ?
  - Length Normalization
  - Importance Sampling
- Results and Analysis

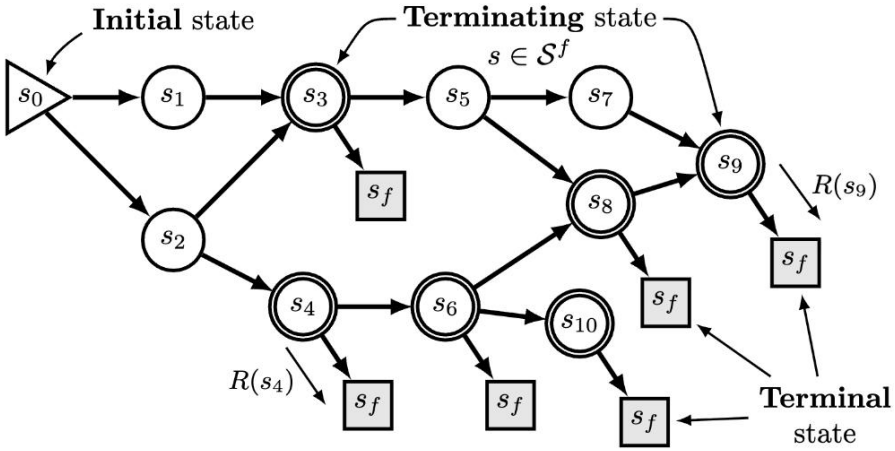
Preliminaries 1: REINFORCE, PPO, GRPO



Shao Z, Wang P, Zhu Q, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models[J]. arXiv preprint arXiv:2402.03300, 2024.

Preliminaries 2: GFlowNets





$$Z_\theta \prod_{t=1}^{n} P_F(s_t \mid s_{t-1}; \theta) = R(x) \prod_{t=1}^{n} P_B(s_{t-1} \mid s_t; \theta).$$

Bengio Y, Lahlou S, Deleu T, et al. Gflownet foundations[J]. Journal of Machine Learning Research, 2023, 24(210): 1-55.

How to Shift Reward Maximization to Distribution Matching ?

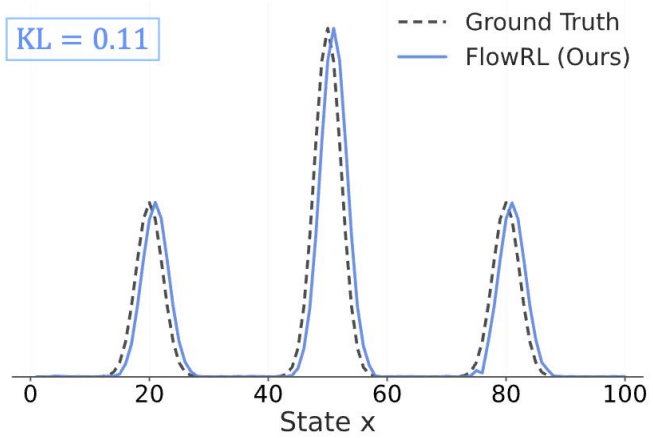
Flow RL: Distribution-matching

$$\min_{\theta} KL(\pi_\theta(\tau) \,||\, \frac{R(\tau)}{Z_\varphi(s_0)})$$

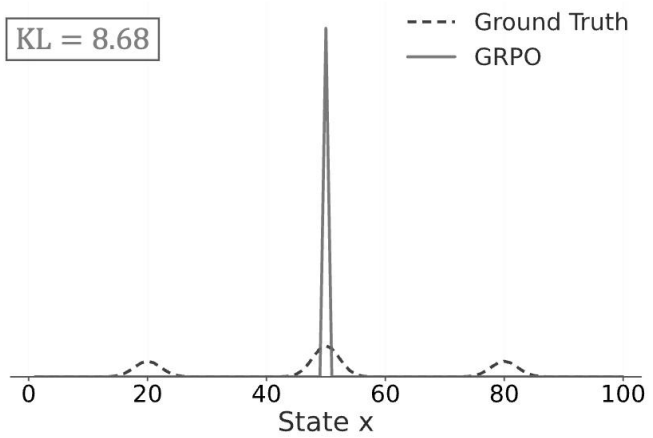
REINFORCE,PPO and GRPO: Reward-maximizing

$$\max_{\theta} \mathbb{E}_{\tau\sim\pi_\theta} R(\tau)$$

Distribution-matching: FlowRL



Reward-maximizing： R++, PPO and GRPO



Trajectory balance as a practical surrogate for KL minimization

$$\min_\theta \mathcal{D}_{\mathrm{KL}}\left(\pi_\theta(\mathbf{y} \mid \mathbf{x}) \,\middle\|\, \frac{\exp(\beta r(\mathbf{x}, \mathbf{y}))}{Z_\phi(\mathbf{x})}\right) \iff \min_\theta \underbrace{\left(\log Z_\phi(\mathbf{x}) + \log \pi_\theta(\mathbf{y} \mid \mathbf{x}) - \beta r(\mathbf{x}, \mathbf{y})\right)^2}_{\textit{Trajectory Balance}}$$

- High-reward trajectories in proportion to their rewards
- Gradient Equivalence
- Stability of Squared Loss

$$\Rightarrow \quad \pi_\theta(\mathbf{y} \mid \mathbf{x}) \propto \exp(\beta r(\mathbf{x}, \mathbf{y})),$$

There are two learnable modules:
(1) policy model
(2) partition function

Partition Function Z_φ

- ➜ Batch mean of rewards: Kimi-k1.5 (derived from the closed form of KL regularization)
- ➜ Variance Minimization Objective [1] : $\mathcal{L}_V(\tau;\theta) = \left(\Phi(\tau;\theta) - \mathbb{E}_\tau[\Phi(\tau;\theta)]\right)^2,$
- ➜ Learbale Partition Function (GFlowNets)

From the flow perspective: Z_ϕ measures the probability flow from the initial state S_0. Intuitively, it estimates the denominator—the sum of rewards across all possible paths—enabling conversion to a probability distribution via $\frac{r(\mathbf{x},\mathbf{y})}{Z_\phi(\mathbf{x})}$.

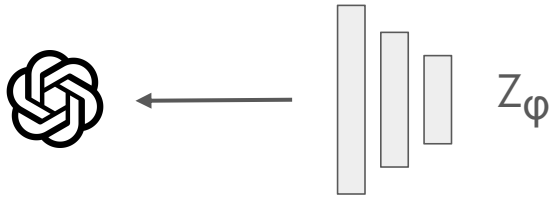
From the implementation perspective: Since the input of Z_ϕ corresponds to the initial state, we utilize the prompt encoded by the language model's final layer hidden states. To convert variable-length prompts to a scalar value, we empirically compute the mean across all tokens.

Team K, Du A, Gao B, et al. Kimi k1. 5: Scaling reinforcement learning with llms[J]. arXiv preprint arXiv:2501.12599, 2025.
Yu F, Jiang L, Kang H, et al. Flow of reasoning: Efficient training of llm policy with divergent thinking[J]. arXiv preprint arXiv:2406.05673, 2024, 1(2): 6.

Implementation of Partition Function Z_φ

- Observation: We found that the above methods for estimating Z are not stable and are quite difficult to learn.

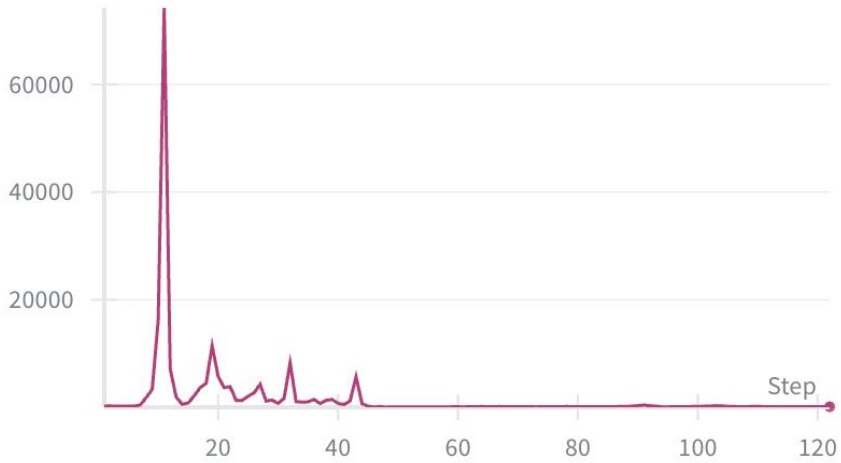
- Key Implementation: Gradients of Z are backpropagated to the policy model



The above objective cannot be scaled up!

- Exploding gradients from long trajectories
- Sampling Mismatch: on / off policy problem

Problem I: Exploding gradients from long trajectories.



$$\log \pi_\theta(\mathbf{y} \mid \mathbf{x})$$

$$\sum_t \log \pi_\theta(\mathbf{y}_t \mid \mathbf{y}_{<t}, \mathbf{x})$$

length normalization

$$\frac{1}{|\mathbf{y}|} \log \pi_\theta(\mathbf{y} \mid \mathbf{x})$$

Problem II: Sampling mismatch.

Trajectory-level importance sampling

$$w = \text{clip}\left(\frac{\pi_\theta(\mathbf{y}|\mathbf{x})}{\pi_{\text{old}}(\mathbf{y}|\mathbf{x})}, 1-\epsilon, 1+\epsilon\right)^{\text{detach}}$$

Note that w is detached (unlike PPO) — corrects distribution shift only, no gradient flow.

Method	AIME 2024	AIME 2025	AMC 2023	MATH-500	Minerva	Olympiad	Avg
FlowRL	15.41	10.83	54.53	66.96	31.41	34.61	35.63
w/o IS	6.25	7.91	41.40	56.97	22.19	25.52	26.71
Zhang et al. [2025a]	10.41	6.66	53.75	66.50	30.97	33.72	33.67

Table 3 | Ablation study on FlowRL with Qwen2.5-7B as the base model. Avg@16 accuracy is reported across six math reasoning benchmarks. IS denotes importance sampling.

FlowRL

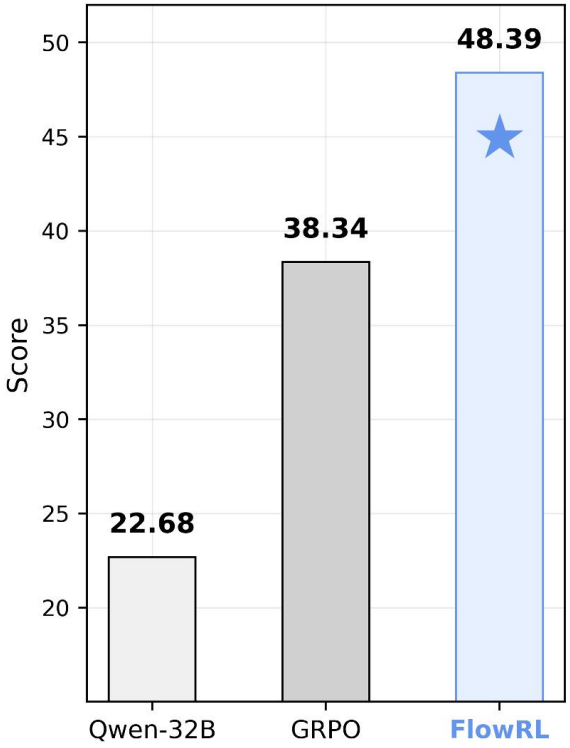
FlowRL

$$\mathcal{L}_{\text{FlowRL}} = w \cdot \left(\log Z_\phi(\mathbf{x}) + \frac{1}{|\mathbf{y}|} \log \pi_\theta(\mathbf{y} \mid \mathbf{x}) - \beta \hat{r}(\mathbf{x}, \mathbf{y}) - \frac{1}{|\mathbf{y}|} \log \pi_{\text{ref}}(\mathbf{y} \mid \mathbf{x}) \right)^2 \tag{6}$$

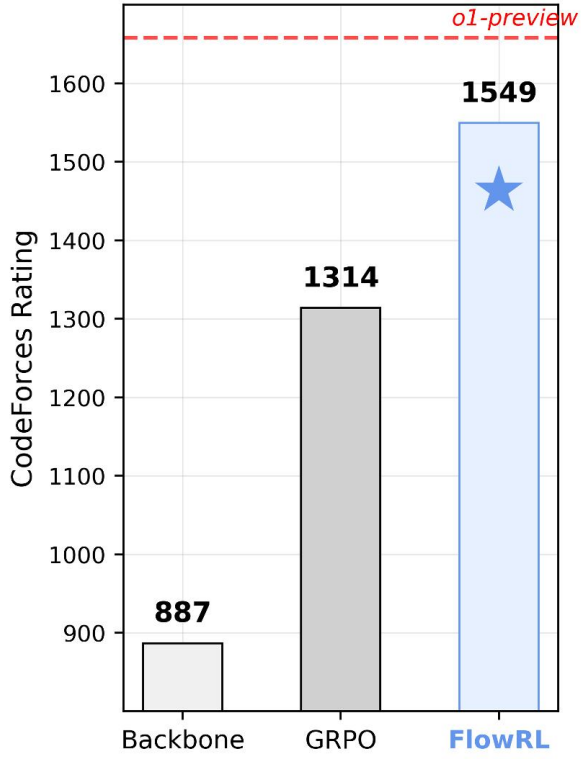
where the clipped importance weight w and normalized reward $\hat{r}(\mathbf{x}, \mathbf{y})$ are defined as:

$$w = \text{clip}\left(\frac{\pi_\theta(\mathbf{y} \mid \mathbf{x})}{\pi_{\text{old}}(\mathbf{y} \mid \mathbf{x})}, 1 - \epsilon, 1 + \epsilon\right)^{\text{detach}}, \quad \hat{r}_i = \frac{r_i - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r})}. \tag{7}$$

Main Results



Math Average Score



CodeForces Rating

Diversity Analysis

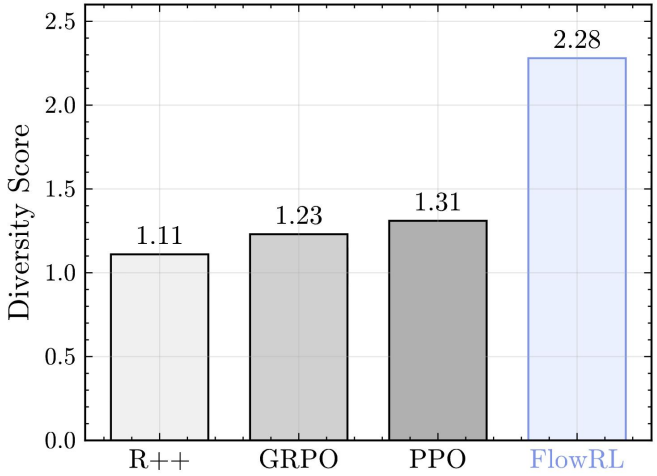


Figure 4 | GPT-judged diversity scores on rollouts of AIME 24/25 problems. FlowRL generates more diverse solutions than R++, GRPO, and PPO.

Case Studies

GRPO exhibits repetitive patterns (AM-GM ×3, identity loops ×2), while FlowRL follows a more diverse solution path.

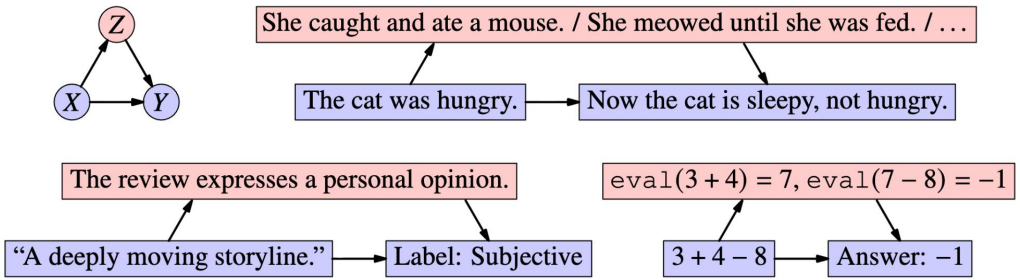
	Content (boxed = actions; "×k" = repeated; "…" = omitted)
Question	Let $\mathcal{B}$ be the set of rectangular boxes with surface area 54 and volume 23. Let r be the radius of the smallest sphere that can contain each box in $\mathcal{B}$. If $r^2 = \frac{p}{q}$ with $\gcd(p, q) = 1$, find $p + q$.
GRPO	"… denote a, b, c … $2(ab+bc+ca) = 54$, $abc = 23$ … $d = \sqrt{a^2 + b^2 + c^2}$, $r = d/2$ … $(a+b+c)^2 = a^2+b^2+c^2 + 2(ab+bc+ca)$ … AM–GM ×3: AM–GM (1) … AM–GM (2) … AM–GM (3) … $(a+b+c)^3$ identity loop ×2: loop (1) … loop (2) … $a = b = c$ (contradiction) … back to $(a+b+c)^2$ … no factorization …"
FlowRL	"… let a, b, c with $2(ab+bc+ca) = 54$, $abc = 23$ … $d = \sqrt{a^2 + b^2 + c^2}$, $r = d/2$ … $(a+b+c)^2 \Rightarrow a^2+b^2+c^2 = s^2 - 54$ … $a = b$ … $a^3 - 27a + 46 = 0$ … rational root $a = 2$ … factor $(a-2)(a^2+2a-23)$ … branch $a = -1 + 2\sqrt{6}$ … back-sub $c = 23/a^2$ … $a^2+b^2+c^2 = \frac{657}{16}$ … $r^2 = \frac{657}{64}$ … Answer 721 …"

Thanks

How to Shift Reward Maximization to Distribution Matching ?

Sampling all potential trajectories is

$$p(Y \mid X) = \sum_Z p_{\mathrm{LM}}(ZY \mid X) = \sum_Z p_{\mathrm{LM}}(Y \mid XZ) p_{\mathrm{LM}}(Z \mid X),$$



Hu E J, Jain M, Elmoznino E, et al. Amortizing intractable inference in large language models[J]. arXiv preprint arXiv:2310.04363, 2023.

How to Shift Reward Maximization to Distribution Matching ?

Core Idea

REINFORCE,PPO and GRPO: Reward-maximizing

$$\max_{\theta} \mathbb{E}_{\tau \sim \pi_\theta} R(\tau)$$

Flow RL: Distribution-matching

$$\min_{\theta} KL(\pi_\theta(\tau) \| \frac{R(\tau)}{Z_\varphi(s_0)})$$

1. Add a partition function to normalize scalar rewards into a target distribution
2. Using reverse KL minimize the distance between the policy and this reward-induced distribution

FlowRL: Flow-Balanced Policy Optimization for LLM Reasoning

